Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Agentic AI for Business Process Automation

TGS Ref No: TGS-2024045801

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2.3

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	31 July 2026	Initial release — 2-day (14-hour) lesson plan for Agentic AI for Business Process Automation, aligned to the 5 hands-on labs and TSC AER-TEM-3026-1.1.	Dr. Alfred Ang
1.1	1 August 2026	Day 2 rescheduled so the final assessment follows instruction within the training day; daily timing now driven from a single source shared with the slide deck; assessment time stated separately from instructional time.	Dr. Alfred Ang
1.2	1 August 2026	Added a screenshot of each lab's application — the four lab websites with their chat widgets open, plus the Lab 1 n8n canvas and sample flyer — to the slide deck and the Learner Guide.	Dr. Alfred Ang
2.0	8 August 2026	Slide deck restyled to the house reference format (visual concept cards, chevron flow strips, comparison panels and takeaway bands). Lab step slides removed from the deck; the click-by-click steps now live only in the Learner Guide, and each lab carries a rendered screenshot of its own n8n canvas. Concept coverage expanded: trigger nodes, data-processing nodes, the four components of an AI agent, instruction design, model parameters, MCP, webhooks (GET/POST, Test vs Production URL) and human-in-the-loop with guardrails.	Dr. Alfred Ang

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2.1	15 October 2026	Topic 1 now opens with the modern view of agentic automation, replacing the separate Foundation section: a brief history of AI 2023 –2026 (prompt, context and harness engineering to AI agents), agentic AI vs AI agents, workflow first vs agent first, AI-assisted workflow creation with n8n Assistant (plus claiming the free AI credits and a social media campaign demo), an n8n training-accounts slide, the five principles of AI security and why workflow first is more secure, and the business value of workflow-first agentic automation. Slide references updated to match.	Dr. Alfred Ang
2.2	15 October 2026	Topic 1 re-sequenced so each lab follows the concepts it needs: n8n fundamentals, then Lab 1, then the AI agent building blocks (AI agent, model, tools and MCP, instructions, memory) and webhooks, then Lab 2. Lab 1 now covers publishing the workflow, copying the form's Production URL and creating a QR-code flyer, with step screenshots and flyers from previous learners. Lab 2 opens with how the building blocks fit together. Platform tiers updated for AI-assisted low-code and AI-generated full-code. New LMS login screenshot; Publish replaces the retired Active toggle. Schedule and slide references updated.	Dr. Alfred Ang

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2.3	16 October 2026	Every lab now follows the knowledge it needs. Topic 2 is taught in two blocks: rule-based vs agentic workflows, designing a safe agentic chatbot and a Telegram HR chatbot demo (BotFather /newbot, built with n8n Assistant from one prompt, then a live chat), then Lab 3; then human in, on and out of the loop, how an approval suspends a workflow, when to put a human in the loop and guardrails, then Lab 4. The agent loop is now explained as Context → Plan → Action → Verify and repeat. Lab workflow files and their n8n names are now 'Lab N - <description>', replacing the old LU/Activity names. Real n8n canvas screenshots added for Lab 2, Lab 3 and the Telegram demo, plus a live Telegram chat. Each lab's key concepts in the Learner Guide now sit directly before that lab. Day 2 schedule and slide references updated.	Dr. Alfred Ang

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Course Information

Course Title	Agentic AI for Business Process Automation
WSQ Course Reference	TGS-2024045801
TSC Title / Code	Artificial Intelligence Application · AER-TEM-3026-1.1
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours)
Daily Timing	Day 1: 9:30 am – 6:30 pm. Day 2: 9:30 am – 6:30 pm instruction, assessment to 8:45 pm. Each day is 8 instructional hours: a 1-hour lunch is excluded, and the two 15-minute tea breaks are counted within the instructional time.
Mode	Instructor-led, hands-on workflow-automation labs built in n8n
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Explain what n8n and AI agents are, and build automated workflows using triggers and actions.
- LO2: Implement AI agent applications that are triggered by webhooks and integrated with business systems.
- LO3: Design agentic chatbot workflows that operate safely within business and regulatory constraints.
- LO4: Apply human-in-the-loop approval so that sensitive actions are authorised by a person.
- LO5: Deploy Retrieval-Augmented Generation (RAG) to ground an agent in an organisation's own documents.

Skills Framework Alignment

This course is mapped to the TSC Artificial Intelligence Application (AER-TEM-3026-1.1).

Code	Ability / Knowledge statement
A1	Implement AI applications in collaboration with technology service providers
A2	Deploy AI workflows according to plan
A3	Identify and report any issues with the AI applications and data collected
K1	Background and basics of AI
K2	Fundamental concepts and methods of statistics

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	and programming for data science
K3	Mathematics and computing theories
K4	Machine learning techniques and applications
K5	Principles of data management
K6	Software development methodologies
K7	Types of AI applications
K8	AI implementation procedures

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on n8n build tasks, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- The Written Assessment covers the Knowledge (K) statements; the Practical Performance covers the Ability (A) statements.
- Final assessment is conducted at the end of Day 2.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — n8n, AI Agents and Agentic Chatbots

Time	Duration	Topic / Activity (with slide references)
9:30–10:00	30 min	Welcome, trainer and learner introductions, ground rules, course outline, learning outcomes, Skills Framework alignment and mandatory digital attendance (AM). Slides 2–19
10:00–11:00	60 min	**Topic 1 — Overview of n8n and AI Agents: why agentic AI, and why workflow first — a brief history of AI 2023–2026, generative AI vs agentic AI, agentic AI vs AI agents, workflow first vs agent first, AI-assisted workflow creation, platform tiers (no-code, AI-assisted low-code, AI-generated full-code), business value, use cases, the five principles of AI security, what n8n is, signing in to n8n, and a demo that builds a social media campaign workflow with n8n Assistant. Slides 20–36**
11:00–11:15	15 min	Tea break
11:15–11:45	30 min	**Topic 1 (continued) — n8n fundamentals: trigger nodes,**

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		data-processing nodes, storage and app actions, JSON and expressions, test vs production executions, and credentials. Slides 37–46
11:45–12:30	45 min	Hands-on Lab 1 — n8n Trigger and Action (Event Flyer with QR Code): build the form-to-Gmail workflow, publish it, copy the form's Production URL and create a QR-code event flyer; present and scan-test the flyers. Slides 47–55
12:30–13:30	60 min	Lunch break
13:30–15:15	105 min	**Topic 1 (continued) — The building blocks of an AI agent, before Lab 2: the AI agent and its loop, the model (popular LLM providers, model parameters and temperature, claiming the free AI credits), tools and MCP, instructions with worked examples, memory, and structured output. Slides 57–69**
15:15–15:30	15 min	Tea break
15:30–16:15	45 min	**Topic 1 (continued) — Webhooks: the inbound URL, webhook vs HTTP Request, GET vs POST, Test vs Production URL and webhook configuration. Slides 70–76**
16:15–18:15	120 min	Hands-on Lab 2 — Webhook and AI Agent (Marina Trust Bank Onboarding): how the webhook, agent, model, tools, instructions, memory and structured output fit together, then build and test the onboarding agent. Slides 77–81
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Instructional time: 480 minutes (8 hours). The day runs 9:30 am - 6:30 pm: the 1-hour lunch is excluded from instructional time and the tea breaks are counted within it.

Day 2 — Human in the Loop, RAG and Assessment

Time	Duration	Topic / Activity (with slide references)
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–10:30	45 min	**Topic 2 — Agentic Chatbot and Human in the Loop, before Lab 3: rule-based vs agentic workflows, designing**

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		a safe agentic chatbot, and a chatbot demo on Telegram — learners create a bot with BotFather /newbot, build an HR policy chatbot with n8n Assistant from one prompt, then connect, publish and chat with it. Slides 82–90
10:30–11:00	30 min	Hands-on Lab 3 — Chatbot Webhook (Investment Advisor): import the workflow, read the contact gate, the non-advisory rule and the FAQ, and deploy the chatbot behind the website. Slides 91–94
11:00–11:15	15 min	Tea break
11:15–11:45	30 min	Hands-on Lab 3 — Chatbot Webhook (continued): test the contact gate and the four compliance probes, and debrief.
11:45–12:30	45 min	**Topic 2 (continued) — Human in the loop and guardrails, before Lab 4: human in, on and out of the loop; how an approval suspends a running workflow; when a human belongs in the workflow; and guardrails. Slides 95–98**
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	Hands-on Lab 4 — Human in the Loop (Client Rapport Assistant): the approve and decline paths. Slides 99–101
15:00–15:15	15 min	Tea break
15:15–16:30	75 min	**Topic 3 — Retrieval-Augmented Generation: why retrieval rather than a bigger prompt, the RAG pipeline, tokenization and embeddings, chunking, vector stores and grounding. Slides 103–111**
16:30–18:00	90 min	Hands-on Lab 5 — RAG Customer Care Chatbot (Cook & Bake Academy): ingestion and the grounded chatbot. Slides 112–115
18:00–18:30	30 min	Course recap, taking it back to work, Q&A. Slides 116–118
18:30–18:45	15 min	**Briefing for Assessment and Assessment digital attendance**
18:45–19:45	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book**
19:45–20:45	60 min	**Practical Performance (PP)**

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		— hands-on n8n build tasks, 1 hour, open book. TRAQOM survey and Assessment digital attendance

Instructional time: 480 minutes (8 hours). Assessment: 2 hours (WA 1 hour + PP 1 hour), plus a 15-minute briefing. The day runs 9:30 am - 6:30 pm: the 1-hour lunch is excluded from instructional time and the tea breaks are counted within it.

Total instructional time across the course: 16 hours (2 days × 8 hours), plus 2 hours of assessment (WA 1 hour + PP 1 hour) and a 15-minute assessment briefing.

Lab Reference (aligned to topics and learning outcomes)

Topic	Outcomes	Labs	Deck slides
Topic 01: Overview of n8n and AI Agents	LO1 · LO2	Lab 1, Lab 2	Slides 20–81
Topic 02: Agentic Chatbot and Human in the Loop	LO3 · LO4	Lab 3, Lab 4	Slides 82–102
Topic 03: Retrieval-Augmented Generation (RAG)	LO5	Lab 5	Slides 103–115

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